and $b = \hat{\mathbf{R}}\cdot\hat{\mathbf{z}} = \cos\theta$. Upon inserting these expressions for a and b in Eq. (3.62) and replacing $\hat{\mathbf{r}}$ with Eq. (3.56a), we have

$$\hat{\mathbf{R}} = \hat{\mathbf{x}}\sin\theta\cos\phi + \hat{\mathbf{y}}\sin\theta\sin\phi + \hat{\mathbf{z}}\cos\theta. \tag{3.64a}$$

A similar procedure can be followed to obtain the expression for $\hat{\boldsymbol{\theta}}$:

$$\hat{\boldsymbol{\theta}} = \hat{\mathbf{x}}\cos\theta\cos\phi + \hat{\mathbf{y}}\cos\theta\sin\phi - \hat{\mathbf{z}}\sin\theta. \tag{3.64b}$$

Finally $\hat{\boldsymbol{\phi}}$ is given by Eq. (3.56b) as

$$\hat{\boldsymbol{\phi}} = -\hat{\mathbf{x}}\sin\phi + \hat{\mathbf{y}}\cos\phi. \tag{3.64c}$$

Equations (3.64a) through (3.64c) can be solved simultaneously to give the expressions for $(\hat{\mathbf{x}},\hat{\mathbf{y}},\hat{\mathbf{z}})$ in terms of $(\hat{\mathbf{R}},\hat{\boldsymbol{\theta}},\hat{\boldsymbol{\phi}})$:

$$\hat{\mathbf{x}} = \hat{\mathbf{R}}\sin\theta\cos\phi + \hat{\boldsymbol{\theta}}\cos\theta\cos\phi - \hat{\boldsymbol{\phi}}\sin\phi, \tag{3.65a}$$
$$\hat{\mathbf{y}} = \hat{\mathbf{R}}\sin\theta\sin\phi + \hat{\boldsymbol{\theta}}\cos\theta\sin\phi + \hat{\boldsymbol{\phi}}\cos\phi, \tag{3.65b}$$
$$\hat{\mathbf{z}} = \hat{\mathbf{R}}\cos\theta - \hat{\boldsymbol{\theta}}\sin\theta. \tag{3.65c}$$

Equations (3.64a) to (3.65c) also can be used to transform Cartesian components (A_x,A_y,A_z) of vector $\mathbf{A}$ into their spherical counterparts (A_R,A_θ,A_ϕ), and vice versa, by replacing $(\hat{\mathbf{x}},\hat{\mathbf{y}},\hat{\mathbf{z}},\hat{\mathbf{R}},\hat{\boldsymbol{\theta}},\hat{\boldsymbol{\phi}})$ with $(A_x,A_y,A_z,A_R,A_\theta,A_\phi)$.

Example 3-9: Cartesian to Spherical Transformation

Express vector $\mathbf{A} = \hat{\mathbf{x}}(x+y) + \hat{\mathbf{y}}(y-x) + \hat{\mathbf{z}}z$ in spherical coordinates.

Solution: Using the transformation relation for A_R given in Table 3-2, we have

$$\begin{aligned} A_R &= A_x\sin\theta\cos\phi + A_y\sin\theta\sin\phi + A_z\cos\theta \\ &= (x+y)\sin\theta\cos\phi + (y-x)\sin\theta\sin\phi + z\cos\theta. \end{aligned}$$

Using the expressions for x, y, and z given by Eq. (3.61c), we have

$$\begin{aligned} A_R &= (R\sin\theta\cos\phi + R\sin\theta\sin\phi)\sin\theta\cos\phi \\ &\quad + (R\sin\theta\sin\phi - R\sin\theta\cos\phi)\sin\theta\sin\phi + R\cos^2\theta \\ &= R\sin^2\theta(\cos^2\phi + \sin^2\phi) + R\cos^2\theta \\ &= R\sin^2\theta + R\cos^2\theta = R. \end{aligned}$$

Similarly,

$$\begin{aligned} A_\theta &= (x+y)\cos\theta\cos\phi + (y-x)\cos\theta\sin\phi - z\sin\theta, \\ A_\phi &= -(x+y)\sin\phi + (y-x)\cos\phi, \end{aligned}$$

and following the procedure used with A_R, we obtain

$$A_\theta = 0, \qquad A_\phi = -R\sin\theta.$$

Hence,

$$\mathbf{A} = \hat{\mathbf{R}}A_R + \hat{\boldsymbol{\theta}}A_\theta + \hat{\boldsymbol{\phi}}A_\phi = \hat{\mathbf{R}}R - \hat{\boldsymbol{\phi}}R\sin\theta.$$

3-3.3 Cylindrical to Spherical Transformations

Transformations between cylindrical and spherical coordinates can be realized by combining the transformation relations of the preceding two subsections. The results are given in Table 3-2.

3-3.4 Distance between Two Points

In Cartesian coordinates, the distance d between two points $P_1 = (x_1,y_1,z_1)$ and $P_2 = (x_2,y_2,z_2)$ is given by Eq. (3.12) as

$$d = |\mathbf{R}_{12}| = [(x_2-x_1)^2 + (y_2-y_1)^2 + (z_2-z_1)^2]^{1/2}. \tag{3.66}$$

Upon using Eq. (3.52) to convert the Cartesian coordinates of P_1 and P_2 into their cylindrical equivalents, we have

$$\begin{aligned} d &= \big[(r_2\cos\phi_2 - r_1\cos\phi_1)^2 \\ &\quad + (r_2\sin\phi_2 - r_1\sin\phi_1)^2 + (z_2-z_1)^2\big]^{1/2} \\ &= \big[r_2^2 + r_1^2 - 2r_1r_2\cos(\phi_2-\phi_1) + (z_2-z_1)^2\big]^{1/2}. \end{aligned} \tag{3.67}$$

(cylindrical)

A similar transformation using Eqs. (3.61a) through (3.61c) leads to an expression for d in terms of the spherical coordinates of P_1 and P_2:

$$\begin{aligned} d &= \big\{R_2^2 + R_1^2 - 2R_1R_2[\cos\theta_2\cos\theta_1 \\ &\quad + \sin\theta_1\sin\theta_2\cos(\phi_2-\phi_1)]\big\}^{1/2}. \end{aligned} \tag{3.68}$$

(spherical)

和 b = R̂ · ẑ = cos θ。在等式中插入 a 和 b 的这些表达式后：（3.62）并用等式替换r̂。（3.56a），我们有

$$\hat{\mathbf{R}} = \hat{\mathbf{x}}\sin\theta\cos\phi + \hat{\mathbf{y}}\sin\theta\sin\phi + \hat{\mathbf{z}}\cos\theta. \tag{3.64a}$$

可以遵循类似的过程来获得 $\hat{\boldsymbol{\theta}}$ 的表达式：

$$\hat{\boldsymbol{\theta}} = \hat{\mathbf{x}}\cos\theta\cos\phi + \hat{\mathbf{y}}\cos\theta\sin\phi - \hat{\mathbf{z}}\sin\theta. \tag{3.64b}$$

最后，$\hat{\boldsymbol{\phi}}$ 由等式给出。（3.56b）为

$$\hat{\boldsymbol{\phi}} = -\hat{\mathbf{x}}\sin\phi + \hat{\mathbf{y}}\cos\phi. \tag{3.64c}$$

方程 (3.64a) 到 (3.64c) 可以同时求解，以 $(\hat{R},\hat{\boldsymbol{\theta}},\hat{\boldsymbol{\phi}})$ 的形式给出 $(\hat{x},\hat{y},\hat{z})$ 的表达式：

$$\hat{\mathbf{x}} = \hat{\mathbf{R}}\sin\theta\cos\phi + \hat{\boldsymbol{\theta}}\cos\theta\cos\phi - \hat{\boldsymbol{\phi}}\sin\phi, \tag{3.65a}$$
$$\hat{\mathbf{y}} = \hat{\mathbf{R}}\sin\theta\sin\phi + \hat{\boldsymbol{\theta}}\cos\theta\sin\phi + \hat{\boldsymbol{\phi}}\cos\phi, \tag{3.65b}$$
$$\hat{\mathbf{z}} = \hat{\mathbf{R}}\cos\theta - \hat{\boldsymbol{\theta}}\sin\theta. \tag{3.65c}$$

方程（3.64a）至（3.65c）也可用于将向量 A 的笛卡尔分量（A_x、A_y、A_z）转换为它们的球面分量（A_R、A_θ、A_ϕ），反之亦然，通过替换（x̂、ŷ、ẑ），R̂，$\hat{\boldsymbol{\theta}}$，$\hat{\boldsymbol{\phi}}$）与（$A_x$，$A_y$，$A_z$，$A_R$，$A_\theta$，$A_\phi$）。

示例 3-9：笛卡尔到球面的变换

在球坐标中表达向量 A = x̂(x + y) + ŷ(y − x) + ẑz。

解：利用表3-2中给出的A_R的变换关系，我们有

$$\begin{aligned} A_R &= A_x\sin\theta\cos\phi + A_y\sin\theta\sin\phi + A_z\cos\theta \\ &= (x+y)\sin\theta\cos\phi + (y-x)\sin\theta\sin\phi + z\cos\theta. \end{aligned}$$

使用方程式给出的 x、y 和 z 表达式。（3.61c），我们有

$$\begin{aligned} A_R &= (R\sin\theta\cos\phi + R\sin\theta\sin\phi)\sin\theta\cos\phi \\ &\quad + (R\sin\theta\sin\phi - R\sin\theta\cos\phi)\sin\theta\sin\phi + R\cos^2\theta \\ &= R\sin^2\theta(\cos^2\phi + \sin^2\phi) + R\cos^2\theta \\ &= R\sin^2\theta + R\cos^2\theta = R. \end{aligned}$$

相似地，

$$\begin{aligned} A_\theta &= (x+y)\cos\theta\cos\phi + (y-x)\cos\theta\sin\phi - z\sin\theta, \\ A_\phi &= -(x+y)\sin\phi + (y-x)\cos\phi, \end{aligned}$$

并按照 A_R 使用的过程，我们得到

$$A_\theta = 0, \qquad A_\phi = -R\sin\theta.$$

因此，

$$\mathbf{A} = \hat{\mathbf{R}}A_R + \hat{\boldsymbol{\theta}}A_\theta + \hat{\boldsymbol{\phi}}A_\phi = \hat{\mathbf{R}}R - \hat{\boldsymbol{\phi}}R\sin\theta.$$

3-3.3 圆柱到球面的变换

结合前两小节的变换关系，可以实现柱坐标和球坐标之间的变换。结果如表3-2所示。

3-3.4 两点之间的距离

在笛卡尔坐标系中，两点 $P_1 = (x_1、y_1、z_1)$ 和 $P_2 = (x_2、y_2、z_2)$ 之间的距离 d 由等式给出：(3.12) 作为

$$d = |\mathbf{R}_{12}| = [(x_2-x_1)^2 + (y_2-y_1)^2 + (z_2-z_1)^2]^{1/2}. \tag{3.66}$$

使用方程式后。(3.52) 将 P_1 和 P_2 的笛卡尔坐标转换为它们的圆柱坐标，我们有

$$\begin{aligned} d &= \big[(r_2\cos\phi_2 - r_1\cos\phi_1)^2 \\ &\quad + (r_2\sin\phi_2 - r_1\sin\phi_1)^2 + (z_2-z_1)^2\big]^{1/2} \\ &= \big[r_2^2 + r_1^2 - 2r_1r_2\cos(\phi_2-\phi_1) + (z_2-z_1)^2\big]^{1/2}. \end{aligned} \tag{3.67}$$

(cylindrical)

使用等式进行类似的转换。(3.61a) 到 (3.61c) 得出 d 用 P_1 和 P_2 球坐标表示的表达式：

$$\begin{aligned} d &= \big\{R_2^2 + R_1^2 - 2R_1R_2[\cos\theta_2\cos\theta_1 \\ &\quad + \sin\theta_1\sin\theta_2\cos(\phi_2-\phi_1)]\big\}^{1/2}. \end{aligned} \tag{3.68}$$

(spherical)

Example 3-10: Vector Component

At a given point in space, vectors **A** and **B** are given in cylindrical coordinates by

$$\mathbf{A} = \hat{\mathbf{r}}2 + \hat{\boldsymbol{\phi}}3 - \hat{\mathbf{z}},$$
$$\mathbf{B} = \hat{\mathbf{r}} + \hat{\mathbf{z}}.$$

Determine (a) the scalar component of **B**, or projection, in the direction of **A**, (b) the vector component of **B** in the direction of **A**, and (c) the vector component of **B** perpendicular to **A**.

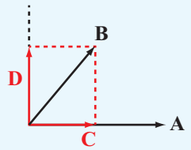


Figure 3-19 Vectors **A**, **B**, **C**, and **D** of Example 3-10.

Solution: **(a)** Let us denote the scalar component of **B** in the direction of **A** as C, as shown in **Fig. 3-19**. Thus,

$$C = \mathbf{B}\cdot\hat{\mathbf{a}} = \mathbf{B}\cdot\frac{\mathbf{A}}{|\mathbf{A}|} = (\hat{\mathbf{r}}+\hat{\mathbf{z}})\cdot\frac{(\hat{\mathbf{r}}2+\hat{\boldsymbol{\phi}}3-\hat{\mathbf{z}})}{\sqrt{4+9+1}} = \frac{2-1}{\sqrt{14}} = 0.267.$$

(b) The vector component of **B** in the direction of **A** is given by the product of the scalar component C and the unit vector $\hat{\mathbf{a}}$:

$$\mathbf{C} = \hat{\mathbf{a}}C = \frac{\mathbf{A}}{|\mathbf{A}|}C = \frac{(\hat{\mathbf{r}}2+\hat{\boldsymbol{\phi}}3-\hat{\mathbf{z}})}{\sqrt{14}}\times 0.267$$
$$= \hat{\mathbf{r}}0.143 + \hat{\boldsymbol{\phi}}0.214 - \hat{\mathbf{z}}0.071.$$

(c) The vector component of **B** perpendicular to **A** is equal to **B** minus **C**:

$$\mathbf{D} = \mathbf{B} - \mathbf{C} = (\hat{\mathbf{r}}+\hat{\mathbf{z}}) - (\hat{\mathbf{r}}0.143 + \hat{\boldsymbol{\phi}}0.214 - \hat{\mathbf{z}}0.071)$$
$$= \hat{\mathbf{r}}0.857 - \hat{\boldsymbol{\phi}}0.214 + \hat{\mathbf{z}}0.929.$$

Concept Question 3-7: Why do we use more than one coordinate system?

Concept Question 3-8: Why is it that the base vectors $(\hat{\mathbf{x}},\hat{\mathbf{y}},\hat{\mathbf{z}})$ are independent of the location of a point, but $\hat{\mathbf{r}}$ and $\hat{\boldsymbol{\phi}}$ are not?

Concept Question 3-9: What are the cyclic relations for the base vectors in (a) Cartesian coordinates, (b) cylindrical coordinates, and (c) spherical coordinates?

Concept Question 3-10: How is the position vector of a point in cylindrical coordinates related to its position vector in spherical coordinates?

Exercise 3-7: Point $P = (2\sqrt{3}, \pi/3, -2)$ is given in cylindrical coordinates. Express P in spherical coordinates.

Answer: $P = (4, 2\pi/3, \pi/3)$. (See EM.)

Exercise 3-8: Transform vector

$$\mathbf{A} = \hat{\mathbf{x}}(x+y) + \hat{\mathbf{y}}(y-x) + \hat{\mathbf{z}}z$$

from Cartesian to cylindrical coordinates.

Answer: $\mathbf{A} = \hat{\mathbf{r}}r - \hat{\boldsymbol{\phi}}r + \hat{\mathbf{z}}z$. (See EM.)

3-4 Gradient of a Scalar Field

When dealing with a scalar physical quantity whose magnitude depends on a single variable, such as the temperature T as a function of height z, the rate of change of T with height can be described by the derivative dT/dz. However, if T is also a function of x and y, its spatial rate of change becomes more difficult to describe because we now have to deal with three separate variables. The differential change in T along x, y, and z can be described in terms of the partial derivatives of T with respect to the three coordinate variables, but it is not immediately obvious as to how we should combine the three partial derivatives so as to describe the spatial rate of change of T along a specified direction. Furthermore, many of the quantities we deal with in electromagnetics are vectors; therefore, both their magnitudes and directions may vary with spatial position. To this end, we introduce three fundamental operators to describe the differential spatial variations of scalars and vectors: the ***gradient***, ***divergence***, and ***curl*** operators. The gradient operator applies to scalar fields and is the subject of the present section. The other two operators, which apply to vector fields, are discussed in succeeding sections.

3-4 GRADIENT OF A SCALAR FIELD 147

例 3-10：向量分量

在空间中的给定点，向量 A 和 B 在柱坐标中给出：

$$\mathbf{A} = \hat{\mathbf{r}}2 + \hat{\boldsymbol{\phi}}3 - \hat{\mathbf{z}},$$
$$\mathbf{B} = \hat{\mathbf{r}} + \hat{\mathbf{z}}.$$

确定 (a) B 在 A 方向上的标量分量或投影，(b) B 在 A 方向上的矢量分量，以及 (c) B 垂直于 A 的矢量分量。

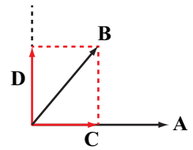


图 3-19 例 3-10 中的向量 A、B、C 和 D。

解：(a) 将 B 在 A 方向上的标量分量表示为 C，如图 3-19 所示。因此，

$$C = \mathbf{B}\cdot\hat{\mathbf{a}} = \mathbf{B}\cdot\frac{\mathbf{A}}{|\mathbf{A}|} = (\hat{\mathbf{r}}+\hat{\mathbf{z}})\cdot\frac{(\hat{\mathbf{r}}2+\hat{\boldsymbol{\phi}}3-\hat{\mathbf{z}})}{\sqrt{4+9+1}} = \frac{2-1}{\sqrt{14}} = 0.267.$$

(b) B 在 A 方向上的向量分量由标量分量 C 和单位向量 â 的乘积给出：

$$\mathbf{C} = \hat{\mathbf{a}}C = \frac{\mathbf{A}}{|\mathbf{A}|}C = \frac{(\hat{\mathbf{r}}2+\hat{\boldsymbol{\phi}}3-\hat{\mathbf{z}})}{\sqrt{14}}\times 0.267$$
$$= \hat{\mathbf{r}}0.143 + \hat{\boldsymbol{\phi}}0.214 - \hat{\mathbf{z}}0.071.$$

(c) B 垂直于 A 的矢量分量等于 B 减去 C：

$$\mathbf{D} = \mathbf{B} - \mathbf{C} = (\hat{\mathbf{r}}+\hat{\mathbf{z}}) - (\hat{\mathbf{r}}0.143 + \hat{\boldsymbol{\phi}}0.214 - \hat{\mathbf{z}}0.071)$$
$$= \hat{\mathbf{r}}0.857 - \hat{\boldsymbol{\phi}}0.214 + \hat{\mathbf{z}}0.929.$$

概念问题 3-7：为什么我们使用多个坐标系？

Concept Question 3-8: Why is it that the base vectors $(\hat{\mathbf{x}},\hat{\mathbf{y}},\hat{\mathbf{z}})$ are independent of the location of a point, but $\hat{\mathbf{r}}$ and $\hat{\boldsymbol{\phi}}$ are not?

Concept Question 3-9: What are the cyclic relations for the base vectors in (a) Cartesian coordinates, (b) cylindrical coordinates, and (c) spherical coordinates?

Concept Question 3-10: How is the position vector of a point in cylindrical coordinates related to its position vector in spherical coordinates?

Exercise 3-7: Point $P = (2\sqrt{3}, \pi/3, -2)$ is given in cylindrical coordinates. Express P in spherical coordinates.

Answer: $P = (4, 2\pi/3, \pi/3)$. (See EM.)

Exercise 3-8: Transform vector

$$\mathbf{A} = \hat{\mathbf{x}}(x+y) + \hat{\mathbf{y}}(y-x) + \hat{\mathbf{z}}z$$

from Cartesian to cylindrical coordinates.

Answer: $\mathbf{A} = \hat{\mathbf{r}}r - \hat{\boldsymbol{\phi}}r + \hat{\mathbf{z}}z$. (See EM.)

3-4 标量场的梯度

当处理大小取决于单个变量的标量物理量时，例如温度 T 作为高度 z 的函数，T 随高度的变化率可以用导数 dT/dz 来描述。然而，如果 T 也是 x 和 y 的函数，则其空间变化率变得更难以描述，因为我们现在必须处理三个单独的变量。 T 沿 x、y 和 z 的微分变化可以用 T 对三个坐标变量的偏导数来描述，但如何结合这三个偏导数来描述 T 沿特定方向的空间变化率并不是立即显而易见的。此外，我们在电磁学中处理的许多量都是矢量。因此，它们的大小和方向都可能随空间位置而变化。为此，我们引入了三个基本算子来描述标量和向量的微分空间变化：梯度、散度和旋度算子。梯度算子适用于标量场，是本节的主题。其他两个适用于向量域的运算符将在后续部分中讨论。

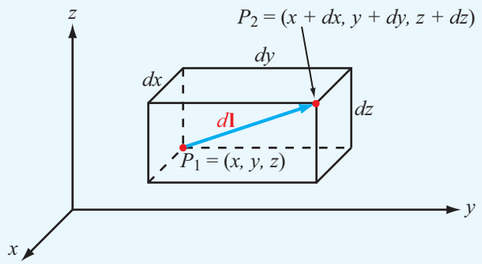


Figure 3-20 Differential distance vector $d\mathbf{l}$ between points P_1 and P_2.

Suppose that $T_1 = T(x,y,z)$ is the temperature at point $P_1 = (x,y,z)$ in some region of space, and

$$T_2 = T(x+dx, y+dy, z+dz)$$

is the temperature at a nearby point $P_2 = (x+dx, y+dy, z+dz)$ (**Fig. 3-20**). The differential distances dx, dy, and dz are the components of the differential distance vector $d\mathbf{l}$. That is,

$$d\mathbf{l} = \hat{\mathbf{x}}\,dx + \hat{\mathbf{y}}\,dy + \hat{\mathbf{z}}\,dz. \tag{3.69}$$

From differential calculus, the temperature difference between points P_1 and P_2, $dT = T_2 - T_1$, is

$$dT = \frac{\partial T}{\partial x}\,dx + \frac{\partial T}{\partial y}\,dy + \frac{\partial T}{\partial z}\,dz. \tag{3.70}$$

Because $dx = \hat{\mathbf{x}} \cdot d\mathbf{l}$, $dy = \hat{\mathbf{y}} \cdot d\mathbf{l}$, and $dz = \hat{\mathbf{z}} \cdot d\mathbf{l}$, Eq. (3.70) can be rewritten as

$$\begin{aligned} dT &= \hat{\mathbf{x}}\frac{\partial T}{\partial x} \cdot d\mathbf{l} + \hat{\mathbf{y}}\frac{\partial T}{\partial y} \cdot d\mathbf{l} + \hat{\mathbf{z}}\frac{\partial T}{\partial z} \cdot d\mathbf{l} \\ &= \left[\hat{\mathbf{x}}\frac{\partial T}{\partial x} + \hat{\mathbf{y}}\frac{\partial T}{\partial y} + \hat{\mathbf{z}}\frac{\partial T}{\partial z}\right] \cdot d\mathbf{l}. \end{aligned} \tag{3.71}$$

The vector inside the square brackets in Eq. (3.71) relates the change in temperature dT to a vector change in direction $d\mathbf{l}$. This vector is called the ***gradient*** of T (or ***grad*** T for short) and denoted ∇T:

$$\nabla T = \text{grad}\ T = \hat{\mathbf{x}}\frac{\partial T}{\partial x} + \hat{\mathbf{y}}\frac{\partial T}{\partial y} + \hat{\mathbf{z}}\frac{\partial T}{\partial z}. \tag{3.72}$$

Equation (3.71) then can be expressed as

$$dT = \nabla T \cdot d\mathbf{l}. \tag{3.73}$$

The symbol ∇ is called the ***del*** or ***gradient operator*** and is defined as

$$\nabla = \hat{\mathbf{x}}\frac{\partial}{\partial x} + \hat{\mathbf{y}}\frac{\partial}{\partial y} + \hat{\mathbf{z}}\frac{\partial}{\partial z} \qquad \textbf{(Cartesian)}. \tag{3.74}$$

▶ Whereas the gradient operator itself has no physical meaning, it attains a physical meaning once it operates on a scalar quantity, and the result of the operation is a vector with magnitude equal to the maximum rate of change of the physical quantity per unit distance and pointing in the direction of maximum increase. ◀

With $d\mathbf{l} = \hat{\mathbf{a}}_l dl$, where $\hat{\mathbf{a}}_l$ is the unit vector of $d\mathbf{l}$, the ***directional derivative*** of T along $\hat{\mathbf{a}}_l$ is

$$\frac{dT}{dl} = \nabla T \cdot \hat{\mathbf{a}}_l. \tag{3.75}$$

We can find the difference $(T_2 - T_1)$, where $T_1 = T(x_1,y_1,z_1)$ and $T_2 = T(x_2,y_2,z_2)$ are the values of T at points

$$P_1 = (x_1,y_1,z_1) \qquad \text{and} \qquad P_2 = (x_2,y_2,z_2)$$

not necessarily infinitesimally close to one another, by integrating both sides of Eq. (3.73). Thus,

$$T_2 - T_1 = \int_{P_1}^{P_2} \nabla T \cdot d\mathbf{l}. \tag{3.76}$$

Example 3-11: Directional Derivative

Find the directional derivative of $T = x^2 + y^2 z$ along direction $\hat{\mathbf{x}}2 + \hat{\mathbf{y}}3 - \hat{\mathbf{z}}2$ and evaluate it at $(1, -1, 2)$.

Solution: First, we find the gradient of T:

$$\nabla T = \left(\hat{\mathbf{x}}\frac{\partial}{\partial x} + \hat{\mathbf{y}}\frac{\partial}{\partial y} + \hat{\mathbf{z}}\frac{\partial}{\partial z}\right)(x^2 + y^2 z) = \hat{\mathbf{x}}2x + \hat{\mathbf{y}}2yz + \hat{\mathbf{z}}y^2.$$

We denote $\mathbf{l}$ as the given direction,

$$\mathbf{l} = \hat{\mathbf{x}}2 + \hat{\mathbf{y}}3 - \hat{\mathbf{z}}2.$$

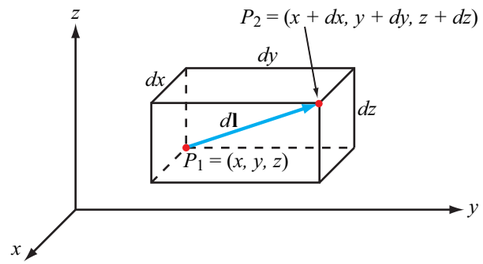


图 3-20 点 P_1 和 P_2 之间的差分距离向量 dl。

假设 $T_1 = T(x, y, z)$ 是空间某个区域中 $P_1 = (x, y, z)$ 点的温度，并且

$$T_2 = T(x+dx, y+dy, z+dz)$$

是附近点 $P_2 = (x+dx, y+dy, z+dz)$ (的温度3-20)。差分距离dx、dy、dz是差分距离矢量dl的分量。那是，

$$d\mathbf{l} = \hat{\mathbf{x}}\,dx + \hat{\mathbf{y}}\,dy + \hat{\mathbf{z}}\,dz. \tag{3.69}$$

根据微分学，点 P_1 和 P_2 之间的温度差 $dT = T_2 - T_1$ 为

$$dT = \frac{\partial T}{\partial x}\,dx + \frac{\partial T}{\partial y}\,dy + \frac{\partial T}{\partial z}\,dz. \tag{3.70}$$

因为 $dx = \hat{x} \cdot dl$、$dy = \hat{y} \cdot dl$ 和 $dz = \hat{z} \cdot dl$，等式：(3.70) 可以重写为

$$\begin{aligned} dT &= \hat{\mathbf{x}}\frac{\partial T}{\partial x} \cdot d\mathbf{l} + \hat{\mathbf{y}}\frac{\partial T}{\partial y} \cdot d\mathbf{l} + \hat{\mathbf{z}}\frac{\partial T}{\partial z} \cdot d\mathbf{l} \\ &= \left[\hat{\mathbf{x}}\frac{\partial T}{\partial x} + \hat{\mathbf{y}}\frac{\partial T}{\partial y} + \hat{\mathbf{z}}\frac{\partial T}{\partial z}\right] \cdot d\mathbf{l}. \end{aligned} \tag{3.71}$$

方程中方括号内的向量。(3.71) 将温度 dT 的变化与方向 dl 上的矢量变化联系起来。该向量称为 T 的梯度（或简称 grad T），记为 ∇T：

$$\nabla T = \text{grad}\ T = \hat{\mathbf{x}}\frac{\partial T}{\partial x} + \hat{\mathbf{y}}\frac{\partial T}{\partial y} + \hat{\mathbf{z}}\frac{\partial T}{\partial z}. \tag{3.72}$$

则方程(3.71)可表示为

$$dT = \nabla T \cdot d\mathbf{l}. \tag{3.73}$$

符号 ∇ 称为 del 或梯度运算符，定义为

$$\nabla = \hat{\mathbf{x}}\frac{\partial}{\partial x} + \hat{\mathbf{y}}\frac{\partial}{\partial y} + \hat{\mathbf{z}}\frac{\partial}{\partial z} \qquad \textbf{(Cartesian)}. \tag{3.74}$$

▶ 梯度算子本身没有物理意义，但它对标量进行运算后就具有物理意义，运算结果是一个向量，其大小等于该物理量单位距离的最大变化率，并指向最大增加的方向。 ◀

对于 $d\mathbf{l} = \hat{\mathbf{a}}_l dl$，其中 $\hat{\mathbf{a}}_l$ 是 dl 的单位向量，T 沿 $\hat{\mathbf{a}}_l$ 的方向导数为

$$\frac{dT}{dl} = \nabla T \cdot \hat{\mathbf{a}}_l. \tag{3.75}$$

我们可以求出差值 $(T_2 - T_1)$，其中 $T_1 = T(x_1, y_1, z_1)$ 和 $T_2 = T(x_2, y_2, z_2)$ 是 T 在点处的值

$$P_1 = (x_1,y_1,z_1) \qquad \text{and} \qquad P_2 = (x_2,y_2,z_2)$$

通过对方程两边求积分，不一定彼此无限小地接近。 (3.73) 。因此，

$$T_2 - T_1 = \int_{P_1}^{P_2} \nabla T \cdot d\mathbf{l}. \tag{3.76}$$

例 3-11：方向导数

求 $T = x^2 + y^2 z$ 沿方向 $\hat{\mathbf{x}}2 + \hat{\mathbf{y}}3 - \hat{\mathbf{z}}2$ 的方向导数，并在 $(1, -1, 2)$ 处对其求值。

解决方案：首先，我们求 T 的梯度：

$$\nabla T = \left(\hat{\mathbf{x}}\frac{\partial}{\partial x} + \hat{\mathbf{y}}\frac{\partial}{\partial y} + \hat{\mathbf{z}}\frac{\partial}{\partial z}\right)(x^2 + y^2 z) = \hat{\mathbf{x}}2x + \hat{\mathbf{y}}2yz + \hat{\mathbf{z}}y^2.$$

我们将 l 表示为给定方向，

$$\mathbf{l} = \hat{\mathbf{x}}2 + \hat{\mathbf{y}}3 - \hat{\mathbf{z}}2.$$

Its unit vector is

$$\hat{\mathbf{a}}_l = \frac{\mathbf{l}}{|\mathbf{l}|} = \frac{\hat{\mathbf{x}}2 + \hat{\mathbf{y}}3 - \hat{\mathbf{z}}2}{\sqrt{2^2+3^2+2^2}} = \frac{\hat{\mathbf{x}}2 + \hat{\mathbf{y}}3 - \hat{\mathbf{z}}2}{\sqrt{17}}.$$

Application of Eq. (3.75) gives

$$\frac{dT}{dl} = \nabla T \cdot \hat{\mathbf{a}}_l = (\hat{\mathbf{x}}2x + \hat{\mathbf{y}}2yz + \hat{\mathbf{z}}y^2) \cdot \left(\frac{\hat{\mathbf{x}}2 + \hat{\mathbf{y}}3 - \hat{\mathbf{z}}2}{\sqrt{17}}\right)$$
$$= \frac{4x + 6yz - 2y^2}{\sqrt{17}}.$$

At $(1,-1,2)$,

$$\left.\frac{dT}{dl}\right|_{(1,-1,2)} = \frac{4-12-2}{\sqrt{17}} = \frac{-10}{\sqrt{17}}.$$

3-4.1 Gradient Operator in Cylindrical and Spherical Coordinates

Even though Eq. (3.73) was derived using Cartesian coordinates, it should have counterparts in other coordinate systems. To convert Eq. (3.72) into cylindrical coordinates (r,ϕ,z), we start by restating the coordinate relations

$$r = \sqrt{x^2+y^2}, \qquad \tan\phi = \frac{y}{x}. \tag{3.77}$$

From differential calculus,

$$\frac{\partial T}{\partial x} = \frac{\partial T}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial T}{\partial \phi}\frac{\partial \phi}{\partial x} + \frac{\partial T}{\partial z}\frac{\partial z}{\partial x}. \tag{3.78}$$

Since z is orthogonal to x and $\partial z/\partial x = 0$, the last term in Eq. (3.78) vanishes. From the coordinate relations given by Eq. (3.77), it follows that

$$\frac{\partial r}{\partial x} = \frac{x}{\sqrt{x^2+y^2}} = \cos\phi, \tag{3.79a}$$

$$\frac{\partial \phi}{\partial x} = -\frac{1}{r}\sin\phi. \tag{3.79b}$$

Hence,

$$\frac{\partial T}{\partial x} = \cos\phi\frac{\partial T}{\partial r} - \frac{\sin\phi}{r}\frac{\partial T}{\partial \phi}. \tag{3.80}$$

This expression can be used to replace the coefficient of $\hat{\mathbf{x}}$ in Eq. (3.72), and a similar procedure can be followed to obtain an expression for $\partial T/\partial y$ in terms of r and ϕ. If, in addition, we use the relations $\hat{\mathbf{x}} = \hat{\mathbf{r}}\cos\phi - \hat{\boldsymbol{\phi}}\sin\phi$ and $\hat{\mathbf{y}} = \hat{\mathbf{r}}\sin\phi + \hat{\boldsymbol{\phi}}\cos\phi$ [from Eqs. (3.57a) and (3.57b)], then Eq. (3.72) becomes

$$\nabla T = \hat{\mathbf{r}}\frac{\partial T}{\partial r} + \hat{\boldsymbol{\phi}}\frac{1}{r}\frac{\partial T}{\partial \phi} + \hat{\mathbf{z}}\frac{\partial T}{\partial z}. \tag{3.81}$$

Hence, the gradient operator in cylindrical coordinates can be expressed as

$$\nabla = \hat{\mathbf{r}}\frac{\partial}{\partial r} + \hat{\boldsymbol{\phi}}\frac{1}{r}\frac{\partial}{\partial \phi} + \hat{\mathbf{z}}\frac{\partial}{\partial z}, \qquad \textbf{(cylindrical)} \tag{3.82}$$

A similar procedure leads to the expression for the gradient in spherical coordinates:

$$\nabla = \hat{\mathbf{R}}\frac{\partial}{\partial R} + \hat{\boldsymbol{\theta}}\frac{1}{R}\frac{\partial}{\partial \theta} + \hat{\boldsymbol{\phi}}\frac{1}{R\sin\theta}\frac{\partial}{\partial \phi}. \qquad \textbf{(spherical)} \tag{3.83}$$

3-4.2 Properties of the Gradient Operator

For any two scalar functions U and V, the following relations apply:

$$(1)\quad \nabla(U+V) = \nabla U + \nabla V, \tag{3.84a}$$

$$(2)\quad \nabla(UV) = U\nabla V + V\nabla U, \tag{3.84b}$$

$$(3)\quad \nabla V^n = nV^{n-1}\nabla V, \qquad \text{for any } n. \tag{3.84c}$$

Example 3-12: Calculating the Gradient

Find the gradient of each of the following scalar functions and then evaluate it at the given point.

(a) $V_1 = 24V_0\cos(\pi y/3)\sin(2\pi z/3)$ at $(3,2,1)$ in Cartesian coordinates,
(b) $V_2 = V_0 e^{-2r}\sin 3\phi$ at $(1,\pi/2,3)$ in cylindrical coordinates,
(c) $V_3 = V_0(a/R)\cos 2\theta$ at $(2a,0,\pi)$ in spherical coordinates.

Solution: (a) Using Eq. (3.72) for ∇,

$$\nabla V_1 = \hat{\mathbf{x}}\frac{\partial V_1}{\partial x} + \hat{\mathbf{y}}\frac{\partial V_1}{\partial y} + \hat{\mathbf{z}}\frac{\partial V_1}{\partial z}$$
$$= -\hat{\mathbf{y}}8\pi V_0\sin\frac{\pi y}{3}\sin\frac{2\pi z}{3} + \hat{\mathbf{z}}16\pi V_0\cos\frac{\pi y}{3}\cos\frac{2\pi z}{3}$$
$$= 8\pi V_0\left[-\hat{\mathbf{y}}\sin\frac{\pi y}{3}\sin\frac{2\pi z}{3} + \hat{\mathbf{z}}2\cos\frac{\pi y}{3}\cos\frac{2\pi z}{3}\right].$$

At $(3,2,1)$,

$$\nabla V_1 = 8\pi V_0\left[-\hat{\mathbf{y}}\sin^2\frac{2\pi}{3} + \hat{\mathbf{z}}2\cos^2\frac{2\pi}{3}\right] = \pi V_0[-\hat{\mathbf{y}}6 + \hat{\mathbf{z}}4].$$

3-4 GRADIENT OF A SCALAR FIELD 149

其单位向量为

$$\hat{\mathbf{a}}_l = \frac{\mathbf{l}}{|\mathbf{l}|} = \frac{\hat{\mathbf{x}}2 + \hat{\mathbf{y}}3 - \hat{\mathbf{z}}2}{\sqrt{2^2+3^2+2^2}} = \frac{\hat{\mathbf{x}}2 + \hat{\mathbf{y}}3 - \hat{\mathbf{z}}2}{\sqrt{17}}.$$

方程的应用(3.75) 给出

$$\frac{dT}{dl} = \nabla T \cdot \hat{\mathbf{a}}_l = (\hat{\mathbf{x}}2x + \hat{\mathbf{y}}2yz + \hat{\mathbf{z}}y^2) \cdot \left(\frac{\hat{\mathbf{x}}2 + \hat{\mathbf{y}}3 - \hat{\mathbf{z}}2}{\sqrt{17}}\right)$$
$$= \frac{4x + 6yz - 2y^2}{\sqrt{17}}.$$

在 (1, −1, 2) 处，

$$\left.\frac{dT}{dl}\right|_{(1,-1,2)} = \frac{4-12-2}{\sqrt{17}} = \frac{-10}{\sqrt{17}}.$$

3-4.1 柱坐标系和球坐标系中的梯度算子

即使方程。 (3.73) 是使用笛卡尔坐标导出的，它在其他坐标系中应该有对应的。转换方程。 (3.72) 转化为柱坐标 (r, ϕ , z)，我们首先重申坐标关系

$$r = \sqrt{x^2+y^2}, \qquad \tan\phi = \frac{y}{x}. \tag{3.77}$$

由微分学可知，

$$\frac{\partial T}{\partial x} = \frac{\partial T}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial T}{\partial \phi}\frac{\partial \phi}{\partial x} + \frac{\partial T}{\partial z}\frac{\partial z}{\partial x}. \tag{3.78}$$

由于 z 与 x 和 ∂ z/∂ x = 0 正交，因此方程中的最后一项(3 .78) 消失。根据方程给出的坐标关系： (3.77) ，由此可知

$$\frac{\partial r}{\partial x} = \frac{x}{\sqrt{x^2+y^2}} = \cos\phi, \tag{3.79a}$$

$$\frac{\partial \phi}{\partial x} = -\frac{1}{r}\sin\phi. \tag{3.79b}$$

因此，

$$\frac{\partial T}{\partial x} = \cos\phi\frac{\partial T}{\partial r} - \frac{\sin\phi}{r}\frac{\partial T}{\partial \phi}. \tag{3.80}$$

该表达式可用于替换式 (1) 中的 x̂ 系数。 (3.72) ，并且可以遵循类似的过程来获得以 r 和 ϕ 表示的 ∂ T /∂ y 的表达式。此外，如果我们使用关系式 x̂ = r̂ cos ϕ − ϕ̂ sin ϕ 和 ŷ = r̂ sin ϕ + ϕ̂ cos ϕ [来自等式： (3.57a) 和 (3.57b)]，则方程(3.72) 变为

$$\nabla T = \hat{\mathbf{r}}\frac{\partial T}{\partial r} + \hat{\boldsymbol{\phi}}\frac{1}{r}\frac{\partial T}{\partial \phi} + \hat{\mathbf{z}}\frac{\partial T}{\partial z}. \tag{3.81}$$

因此，柱坐标系中的梯度算子可以表示为

$$\nabla = \hat{\mathbf{r}}\frac{\partial}{\partial r} + \hat{\boldsymbol{\phi}}\frac{1}{r}\frac{\partial}{\partial \phi} + \hat{\mathbf{z}}\frac{\partial}{\partial z}, \qquad \textbf{(cylindrical)} \tag{3.82}$$

类似的过程可得出球坐标中的梯度表达式：

$$\nabla = \hat{\mathbf{R}}\frac{\partial}{\partial R} + \hat{\boldsymbol{\theta}}\frac{1}{R}\frac{\partial}{\partial \theta} + \hat{\boldsymbol{\phi}}\frac{1}{R\sin\theta}\frac{\partial}{\partial \phi}. \qquad \textbf{(spherical)} \tag{3.83}$$

3-4.2 梯度算子的性质

对于任意两个标量函数 U 和 V ，以下关系适用：

$$(1)\quad \nabla(U+V) = \nabla U + \nabla V, \tag{3.84a}$$

$$(2)\quad \nabla(UV) = U\nabla V + V\nabla U, \tag{3.84b}$$

$$(3)\quad \nabla V^n = nV^{n-1}\nabla V, \qquad \text{for any } n. \tag{3.84c}$$

例 3-12：计算梯度

求以下每个标量函数的梯度，然后在给定点对其求值。

(a) 笛卡尔坐标中 (3, 2, 1) 处的 $V_1 = 24V_0\cos(\pi y/3)\sin(2\pi z/3)$，(b) 圆柱坐标中 $(1, \pi/2, 3)$ 处的 $V_2 = V_0 e^{-2r}\sin 3\phi$ nates，(c) $V_3 = V_0(a/R)\cos 2\theta$ 在球坐标 (2a, 0, π) 处 。

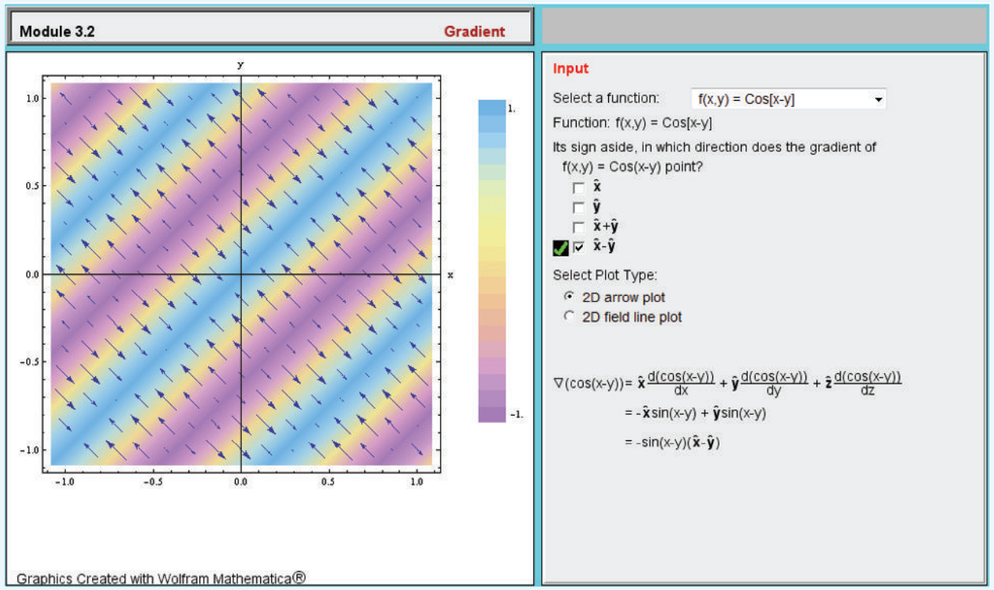
解决方案： (a) 使用等式。 (3.72) 对于 ∇ ，

$$\nabla V_1 = \hat{\mathbf{x}}\frac{\partial V_1}{\partial x} + \hat{\mathbf{y}}\frac{\partial V_1}{\partial y} + \hat{\mathbf{z}}\frac{\partial V_1}{\partial z}$$
$$= -\hat{\mathbf{y}}8\pi V_0\sin\frac{\pi y}{3}\sin\frac{2\pi z}{3} + \hat{\mathbf{z}}16\pi V_0\cos\frac{\pi y}{3}\cos\frac{2\pi z}{3}$$
$$= 8\pi V_0\left[-\hat{\mathbf{y}}\sin\frac{\pi y}{3}\sin\frac{2\pi z}{3} + \hat{\mathbf{z}}2\cos\frac{\pi y}{3}\cos\frac{2\pi z}{3}\right].$$

在 (3, 2, 1) 处，

$$\nabla V_1 = 8\pi V_0\left[-\hat{\mathbf{y}}\sin^2\frac{2\pi}{3} + \hat{\mathbf{z}}2\cos^2\frac{2\pi}{3}\right] = \pi V_0[-\hat{\mathbf{y}}6 + \hat{\mathbf{z}}4].$$

Module 3.2 Gradient Select a scalar function $f(x,y,z)$, evaluate its gradient, and display both in an appropriate 2-D plane.



(b) The function V_2 is expressed in terms of cylindrical variables. Hence, we need to use Eq. (3.82) for ∇:

$$\begin{aligned}\nabla V_2 &= \left(\hat{\mathbf{r}}\frac{\partial}{\partial r}+\hat{\boldsymbol{\phi}}\frac{1}{r}\frac{\partial}{\partial \phi}+\hat{\mathbf{z}}\frac{\partial}{\partial z}\right)V_0 e^{-2r}\sin 3\phi \\ &= -\hat{\mathbf{r}}2V_0 e^{-2r}\sin 3\phi+\hat{\boldsymbol{\phi}}(3V_0 e^{-2r}\cos 3\phi)/r \\ &= \left[-\hat{\mathbf{r}}2\sin 3\phi+\hat{\boldsymbol{\phi}}\frac{3\cos 3\phi}{r}\right]V_0 e^{-2r}.\end{aligned}$$

At $(1,\pi/2,3)$, $r=1$ and $\phi=\pi/2$. Hence,

$$\nabla V_2=\left[-\hat{\mathbf{r}}2\sin\frac{3\pi}{2}+\hat{\boldsymbol{\phi}}3\cos\frac{3\pi}{2}\right]V_0e^{-2}=\hat{\mathbf{r}}2V_0e^{-2}=\hat{\mathbf{r}}0.27V_0.$$

(c) As V_3 is expressed in spherical coordinates, we apply Eq. (3.83) to V_3:

$$\begin{aligned}\nabla V_3 &= \left(\hat{\mathbf{R}}\frac{\partial}{\partial R}+\hat{\boldsymbol{\theta}}\frac{1}{R}\frac{\partial}{\partial\theta}+\hat{\boldsymbol{\phi}}\frac{1}{R\sin\theta}\frac{\partial}{\partial\phi}\right)V_0\left(\frac{a}{R}\right)\cos 2\theta \\ &= -\hat{\mathbf{R}}\frac{V_0a}{R^2}\cos 2\theta-\hat{\boldsymbol{\theta}}\frac{2V_0a}{R^2}\sin 2\theta \\ &= -[\hat{\mathbf{R}}\cos 2\theta+\hat{\boldsymbol{\theta}}2\sin 2\theta]\frac{V_0a}{R^2}.\end{aligned}$$

At $(2a,0,\pi)$, $R=2a$ and $\theta=0$, which yields

$$\nabla V_3=-\hat{\mathbf{R}}\frac{V_0}{4a}.$$

Exercise 3-9: Given $V=x^2y+xy^2+xz^2$, (a) find the gradient of V, and (b) evaluate it at $(1,-1,2)$.

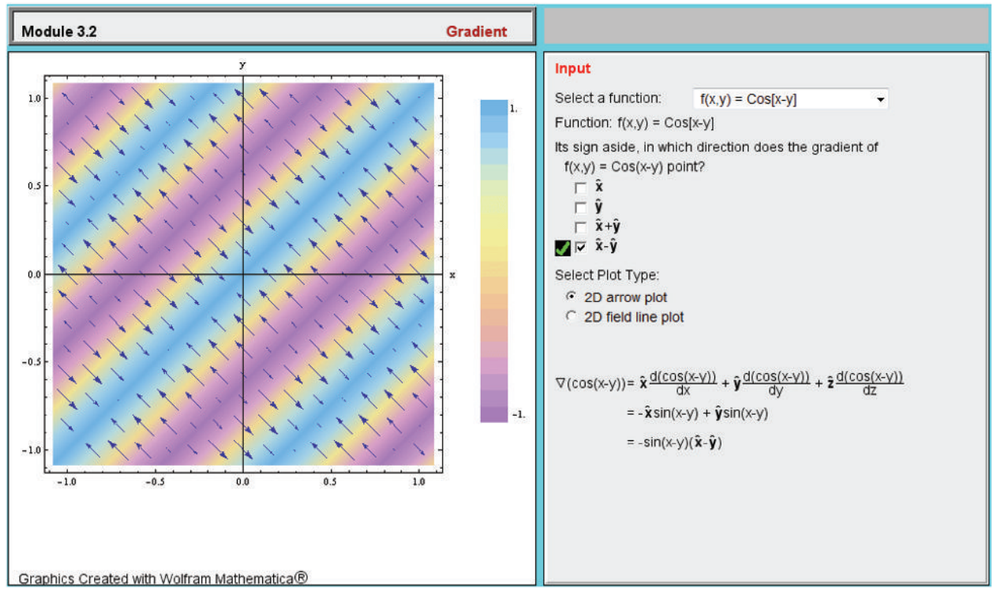
Answer: (a) $\nabla V=\hat{\mathbf{x}}(2xy+y^2+z^2)+\hat{\mathbf{y}}(x^2+2xy)+\hat{\mathbf{z}}2xz$, (b) $\nabla V\big|_{(1,-1,2)}=\hat{\mathbf{x}}3-\hat{\mathbf{y}}+\hat{\mathbf{z}}4$. (See EM.)

Exercise 3-10: Find the directional derivative of $V=rz^2\cos 2\phi$ along the direction of $\mathbf{A}=\hat{\mathbf{r}}2-\hat{\mathbf{z}}$ and evaluate it at $(1,\pi/2,2)$.

Answer: $(dV/dl)\big|_{(1,\pi/2,2)}=-4/\sqrt{5}$. (See EM.)

150 CHAPTER 3 VECTOR ANALYSIS

模块 3.2 梯度 选择标量函数 f (x, y, z)，计算其梯度，并将两者显示在适当的二维平面中。



(b) 函数 V_2 用圆柱变量表示。因此，我们需要使用方程 。(3.82) 对于 ∇：

$$\begin{aligned}\nabla V_2 &= \left(\hat{\mathbf{r}}\frac{\partial}{\partial r}+\hat{\boldsymbol{\phi}}\frac{1}{r}\frac{\partial}{\partial \phi}+\hat{\mathbf{z}}\frac{\partial}{\partial z}\right)V_0 e^{-2r}\sin 3\phi \\ &= -\hat{\mathbf{r}}2V_0 e^{-2r}\sin 3\phi+\hat{\boldsymbol{\phi}}(3V_0 e^{-2r}\cos 3\phi)/r \\ &= \left[-\hat{\mathbf{r}}2\sin 3\phi+\hat{\boldsymbol{\phi}}\frac{3\cos 3\phi}{r}\right]V_0 e^{-2r}.\end{aligned}$$

在 $(1,\pi/2,3)$ 处，$r=1$ 和 $\phi=\pi/2$。因此，

$$\nabla V_2=\left[-\hat{\mathbf{r}}2\sin\frac{3\pi}{2}+\hat{\boldsymbol{\phi}}3\cos\frac{3\pi}{2}\right]V_0e^{-2}=\hat{\mathbf{r}}2V_0e^{-2}=\hat{\mathbf{r}}0.27V_0.$$

(c) 由于 V_3 以球坐标表示，因此我们应用等式： (3.83) 至 V_3：

$$\begin{aligned}\nabla V_3 &= \left(\hat{\mathbf{R}}\frac{\partial}{\partial R}+\hat{\boldsymbol{\theta}}\frac{1}{R}\frac{\partial}{\partial\theta}+\hat{\boldsymbol{\phi}}\frac{1}{R\sin\theta}\frac{\partial}{\partial\phi}\right)V_0\left(\frac{a}{R}\right)\cos 2\theta \\ &= -\hat{\mathbf{R}}\frac{V_0a}{R^2}\cos 2\theta-\hat{\boldsymbol{\theta}}\frac{2V_0a}{R^2}\sin 2\theta \\ &= -[\hat{\mathbf{R}}\cos 2\theta+\hat{\boldsymbol{\theta}}2\sin 2\theta]\frac{V_0a}{R^2}.\end{aligned}$$

在 $(2a, 0, \pi)$ 处，$R=2a$ 和 $\theta=0$，得出

$$\nabla V_3=-\hat{\mathbf{R}}\frac{V_0}{4a}.$$

练习 3-9：给定 $V=x^2y+xy^2+xz^2$，(a) 求 V 的梯度，(b) 在 (1, −1, 2) 处评估它。

答案：(a) $\nabla V=\hat{\mathbf{x}}(2xy+y^2+z^2)+\hat{\mathbf{y}}(x^2+2xy)+\hat{\mathbf{z}}2xz$，(b) $\nabla V\big|_{(1,-1,2)}=\hat{\mathbf{x}}3-\hat{\mathbf{y}}+\hat{\mathbf{z}}4$。（参见EM。）

练习 3-10：求 $V=rz^2\cos 2\phi$ 沿 $\mathbf{A}=\hat{\mathbf{r}}2-\hat{\mathbf{z}}$ 方向的方向导数，并在 $(1,\pi/2,2)$ 处对其求值。

Answer: $(dV/dl)\big|_{(1,\pi/2,2)}=-4/\sqrt{5}$. (See EM.)